

Victoria Junior College

JC2 Preliminary Examination 2009

9747 H2 Biology Paper 3

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Wednesday

1400 – 1530 h

23 September 2009

1Hour 30 Minutes

Name: _____

CT: _____

Instructions

1. This paper has three structured questions and one essay question
2. **Answer ALL questions.**
3. For structured questions, answers must be written on the spaces provided and nowhere else.
For essay question, answers must be written on the writing paper.

Question No.	Marks
1	
2	
3	
4	Essay
Total	

Answer all questions. Write your answers in the space provided and nowhere else.

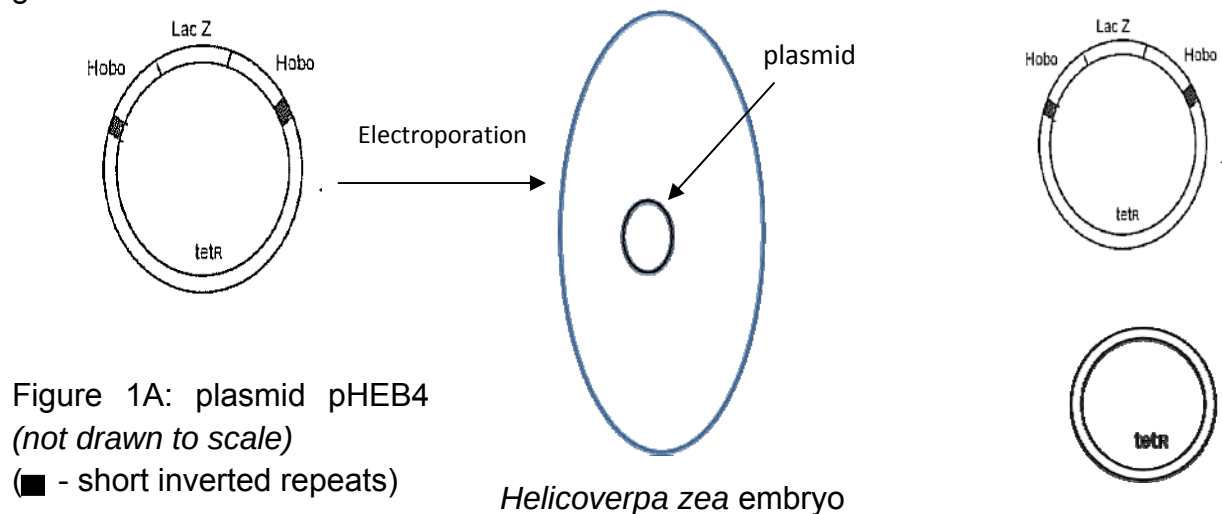
Question 1 (15 marks)

The tobacco budworm (*Heliothis virescens*) and the corn earworm (*Helicoverpa zea*) are pests on a wide variety of crops including cotton, corn, soybean, and other economic and ornamental plants. A study was carried out to assess the possibility of controlling the pest through the insertion of a gene.

In this study, a plasmid pHEB4 containing the *hobo* element from *Drosophila* was used. The *hobo* element is capable of cutting (excision) itself out of the plasmid and then insert into the host genome. The ability of this element to excise itself out is dependent on the short inverted repeats that exist at both ends of the gene.

In order to ascertain the excision of the *hobo* element, a Lac Z gene was inserted into the middle of this element (Figure 1A). The insertion of the Lac Z **does not** affect the ability of the *hobo* element to excise and to insert itself.

The plasmids were then used to transform *H. zea* embryos (Figure 1A). Excision of *hobo* element was analyzed by isolating the plasmids (Figure 1B) from the manipulated embryos (after 22 hours) and used to transform a strain of *E. coli* that lacks the Lac Z gene.



(a) What is the purpose of electroporation? [1]

(b) Using the information from the Figures above as well as your own knowledge,

(i) explain why a strain of bacteria without the Lac Z gene was used in this case. [1]

(ii) describe and explain a procedure (using the Lac Z gene) that would allow one to distinguish between the two types of bacteria. [3]

It was found that the frequency of *hobo* excision varied only slightly from 1.6% - 2.0% of *plasmids* analyzed. The researchers then harvested a new batch of the *H. zea* embryos and treated them exactly like before. However the electroporated embryos were allowed to develop and subsequently these grew into adults. Cells were obtained from the pupal cases and the DNA subjected to PCR. The products were then analyzed to determine the presence of the hobo element. It was found that out of a total of 1000 embryos subjected to electroporation, 17 (or 1.7%) *adults* harbored the *hobo-lacZ* insert.

(c) Explain the importance of PCR in the above experiment. [2]

(d) Outline how the PCR products can be analyzed to allow the identification of the hobo element. [4]

(e) Considering all the information that is available,

(i) suggest a possible replacement for Lac Z in the control of the corn earworm. [1]

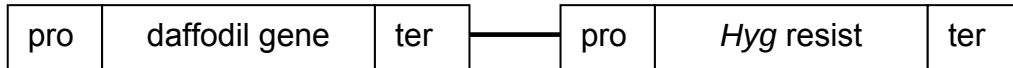
(ii) discuss the feasibility of controlling the corn earworm through this method. [2]

(iii) State one advantage of this form of control. [1]

Question 2 (15 marks)

Rice is the staple diet in many developing parts of the world especially in Asia. However it lacks β -carotene, from which vitamin A is synthesized. Adequate concentrations of vitamin A are needed to give protection from night blindness. Golden rice, which contains β -carotene, was developed by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

Figure 2.1 below shows an artificial DNA sequence used.



Key

pro – start site for polymerase enzymes

ter – end signal for polymerase enzymes

Hyg resist – antibiotic resistance gene from a bacterium

Figure 2.2 shows the main events in obtaining a transgenic plant.

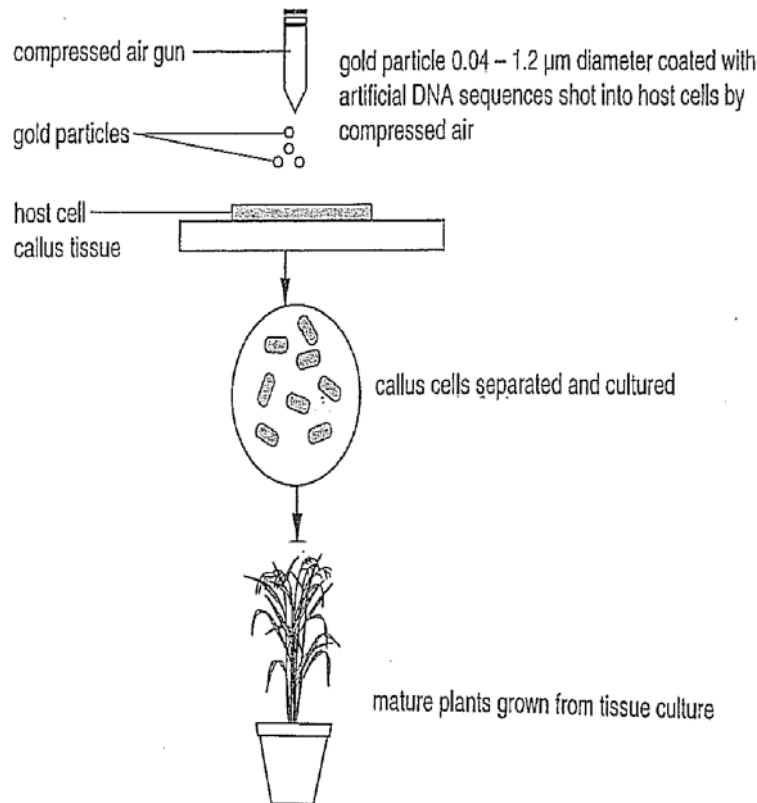


Fig. 2.2

(a) Explain what is meant by 'genetically modifying rice'. [1]

(b) Explain the role of the *Hyg* gene in this procedure. [2]

(c) A complete mature plant can be developed from callus cells. Suggest a reason to account for this phenomenon. [2]

Despite the potential benefits of GM crops, there is concern over the possible environmental and agronomic impacts if the transgenes 'escape' and become established in natural or agricultural ecosystems.

(d) State one problem of transgenes 'escaping' into

(i) natural ecosystem. [1]

(ii) agricultural ecosystem. [1]

Many factors contribute to the ability of the transgenes to escape into the wild relative. Some of these factors include distance between GM and non-GM crops, pollen viability, wind direction and time of planting. **Table 2.1** shows the results obtained from an investigation into factors affecting gene transfer between GM maize (with *Bt* gene) and non-GM maize.

	<i>Same planting date</i>	<i>Up to 10 days difference in planting dates (average 7 days)</i>	<i>More than 10 days planting difference (average 22 days)</i>
Average area of GM (Bt) crop (hectares)	1.28	1.5	6.1
Average distance to 1 st non-GM row (metres)	1	5	2
Average percentage of non-GM grains tested positive for Bt gene in the first row (see data in the row above)	16.93	7	3.56
Average percentage of non-GM grains tested positive for Bt gene in the fourth row (Distance to first row + 2.25 m)	2.73	0.97	0.84
Average percentage of non-GM grains tested positive for Bt gene in the eighth row (Distance to first row + 5.25 m)	1.18	0.63	0.56
Average percentage of non-GM grains tested positive for Bt gene in the sixteenth row (Distance to first row + 11.25 m)	1.02	0.6	0.26

(e) With reference to Table 2.1 above,

- (i) suggest a reason why the size of the GM plot was increased with an increased in the difference of planting time between GM and non-GM plants. [1]

(ii) state the effect of distance (from the GM plants) on gene transfer. Support your answer with data from the Table. [2]

(iii) describe the effect of differences in planting time on gene transfer. [2]

(iv) suggest a reason for this difference. [1]

- (f) Despite concerns over GM crops, researchers believe that these crops can be used to solve the *increasing demand* for food. Use an example to illustrate how a genetically modified crop can be used in this way. [2]

Question 3 (15 marks)

(a) Hematopoietic stem cells are found in the bone marrow of adults, which includes femurs, hip, ribs, sternum, and other bones. The hematopoietic stem cell has been shown to give rise to skin, respiratory epithelium, intestinal epithelium, renal epithelium, liver parenchyma, pancreas, skeletal muscle, vascular endothelium, myocardium, and central nervous system (CNS) neurons.

(i) Discuss the normal function of the “hematopoietic stem cells”. [1]

(ii) Hematopoietic stem cell is a kind of adult stem cell. Distinguish adult stem cells with the zygotic stem cells. [2]

(iii) Explain how it is possible for hematopoietic stem cells to be converted to cell types of a completely different tissue. [1]

(b) In Europe, the commonest mutation of the cystic fibrosis transmembrane conductance regulator (CFTR) gene is $\Delta F508$. Deletion of three base pairs results in the loss of one amino acid, phenylalanine, in the CFTR protein. CFTR with the $\Delta F508$ mutation does not become part of plasma (cell surface) membranes.

A second mutation of CFTR, called R117H, results in the replacement of an arginine amino acid by histidine. CFTR with this mutation does become part of plasma membranes.

Cells, from a certain gland, were taken from three sets of volunteers:

1. homozygous for the normal CFTR allele;
2. heterozygous $\Delta F508$ and R117H;
3. homozygous for the $\Delta F508$ mutation.

The conductance of hydrogencarbonate ions (HCO_3^-) and chloride ions (Cl^-) across the plasma membranes was measured. The results are shown in Figure 3.1.

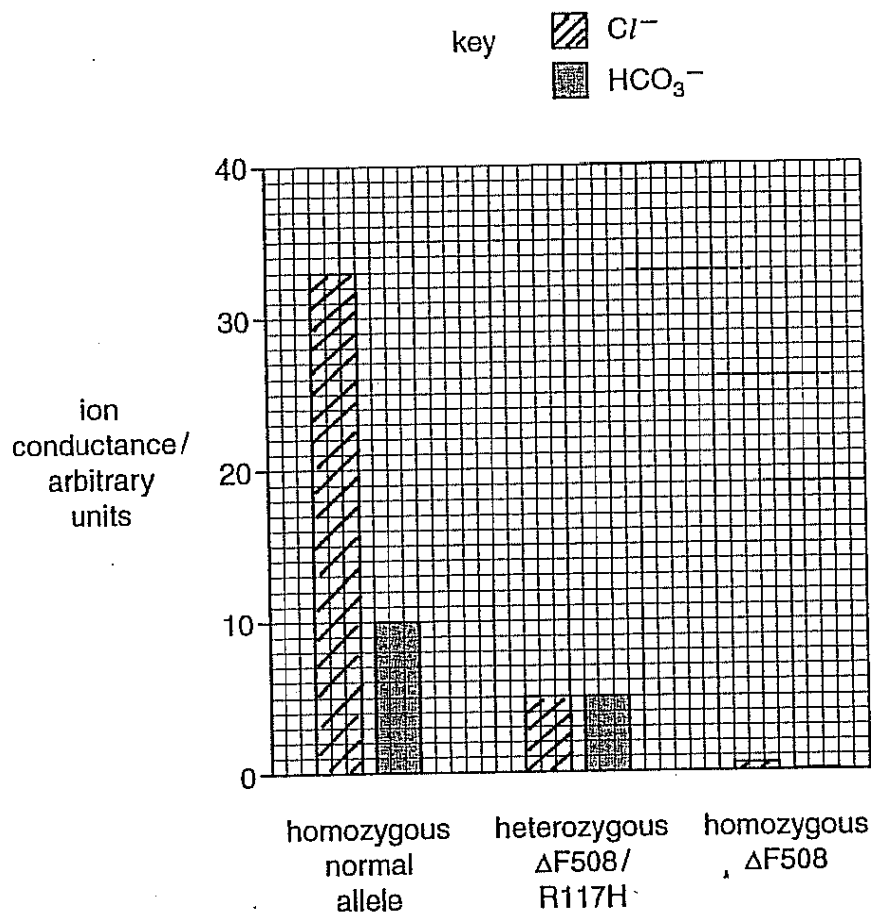


Figure 3.1

- (i) With reference to Figure 3.1 and the information given in the question, explain the different effects of the two CFTR mutations on ion conductance. [4]

- (ii) Calculate the percentage reduction in chloride ion (Cl^-) transport by cells from heterozygous volunteers, in comparison with cells from volunteers homozygous for the normal CFTR allele. Show your working. [1]

- (c) Hypertension is a multigene disease. To overcome hypertension, the antisense somatic gene therapy was introduced. Transfer of the antisense genes to somatic cells was achieved by an *in vivo* approach. It would be possible to try an *ex vivo* approach. However, this strategy has no obvious applicability to hypertension as the cause of the disease lies in the reaction of blood vessels, not in one specific tissue. For the *in vivo* approach, two different strategies had been developed. In the first strategy, the non-viral delivery system was used and it might involve direct injection of naked DNA. Direct injection was effective, but the efficiency of uptake could be further improved. The second strategy involved the use of viral vectors to deliver the antisense DNA.

Use the above information and answer the followings:

- (i) For the treatment of hypertension, the antisense approach was adopted. Explain the basis behind the antisense approach. [2]

- (ii) Both the *in vivo* and *ex vivo* approaches had been used for the transference of the antisense genes into somatic cells. Contrast between the *in-vivo* and the *ex-vivo* approach of gene therapy. [2]

- (iii) In the treatment of hypertension using the *in vivo* approach, liposomes might be preferred, over both direct injection and viral vectors, for the transfer of the antisense genes. In view of the genetic basis of hypertension, suggest a reason why liposomes are preferred. [2]

Write your answers to this question on the separate answer paper provided.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections (a), (b) etc., as indicated in the question.

Question 4 (20 marks)

- a) What are the natural and applied uses of restriction enzymes? [4]
- b) Discuss the uses of RFLP analysis. [9]
- c) Discuss the importance of tissue culture in the production of genetically modified plants. [7]